

04.05.2026 (10.00AM-01.00PM)

VI Sem B.Tech Regular Examinations

Natural Language Processing

Subject Code: 23CACAT6010

Time: 3 Hours

Max. Marks: 70

Instructions:

1. Answer all question from Part-A. Each question carries 2 marks
2. Answer one full question from each unit in Part-B. Each full question carries 10 marks

S.No.	Questions	K	CO	Marks	PO & PSO'S
Session-A					
1	a What are the main challenges in Natural Language Processing?	K2	CO1	[2M]	1, 2& 2
	b What is meant by English Morphology?	K2	CO1	[2M]	1& 2
	c Define Perplexity.	K1	CO2	[2M]	1, 2& 2
	d What is the Objective of Hidden Markov Model.	K1	CO2	[2M]	1, 2& 2
	e What is the purpose of a Treebank in natural language processing?	K1	CO3	[2M]	1, 2& 2
	f Point out chart parsing?	K2	CO3	[2M]	1, 2& 2
	g List the basic elements of First Order logic?	K1	CO4	[2M]	1, 2& 2
	h What are Thematic Roles? Give one example.	K2	CO4	[2M]	1, 2& 2
	i What is the purpose of Lemmatization in NLP?	K2	CO5	[2M]	1, 2& 2
	j Explain the concept of entity-based coherence in discourse.	K2	CO5	[2M]	1, 2& 2
Session-B					
UNIT -I					
2	a What is Klerene closure? Explain with a simple example about Regular expression	K2	CO1	[5M]	1, 2& 2
	b Briefly explain Detecting and correcting and spelling errors	K3	CO1	[5M]	1, 2, 3&2
OR					
3	a Explain how the Minimum Edit Distance is calculated. Also determine the edit distance between the words "INTENTION" and "EXECUTION".	K4	CO1	[5M]	1, 2, 4& 2
	b Generalize the Importance of language modelling	K2	CO1	[5M]	1, 2& 2
UNIT -II					
4	a Explain in detail about the Interpolation and Back-off	K2	CO2	[5M]	1, 2& 2

b	Illustrate about the Log-likelihood function	K2	CO2	[5M]	1, 2& 2
OR					
5	a Discuss the Laplace smoothing	K2	CO2	[5M]	1, 2& 2
	b Examine the various types of word classes?	K4	CO2	[5M]	4& 2
UNIT -III					
6	a Discuss the role of grammar rules in the context of syntactic analysis for English. How do these rules contribute to parsing sentences?	K2	CO3	[5M]	1, 2& 2
	b Demonstrate the basic steps in CYK algorithm and its time complexity	K3	CO3	[5M]	1, 2, 3& 2
OR					
7	a Compose the relationship between syntax and semantics in natural language processing.	K4	CO3	[5M]	4& 2
	b Discuss the applications of shallow parsing in natural language processing.	K2	CO3	[5M]	1, 2& 2
UNIT -IV					
8	a Discuss Syntax-Driven Semantic Analysis with suitable examples.	K4	CO4	[5M]	4& 2
	b What do you mean by word sense disambiguation (WSD)? Discuss dictionary-based approach	K2	CO4	[5M]	1, 2& 2
OR					
9	a Give FOL translations for the following sentences: 1. Vegetarians do not eat meat. 2. Not all vegetarians eat egg.	K3	CO4	[5M]	1, 2, 3& 2
	b What advantages does bootstrapping offer in the context of WSD, and how does it address the data labeling challenge?	K4	CO4	[5M]	4& 2
UNIT -V					
10	a Explain Anaphora Resolution with the help of Hobbs's algorithm.	K2	CO5	[5M]	1, 2& 2
	b Discuss the role of linguistic corpora such as Brown Corpus and British National Corpus (BNC) in language research.	K4	CO5	[5M]	4& 2
OR					
11	a Define discourse segmentation and explain its significance in natural language processing.	K2	CO5	[5M]	1, 2& 2
	b Discuss how Word Net enhances semantic understanding in natural language processing.	K4	CO5	[5M]	4& 2

K* Indicates Blooms Taxonomy Level